

Portland cement

Early civilizations relied on **natural minerals** to bond building materials together. By the 19th century, factories made basic cement from **local limestone and clay**. In 1824, British bricklayer **JOSEPH ASPDIN** ground cooked limestone and clay into powder and **added water**. He named the strong concrete "Portland cement" after quality Portland stone. **AFFORDABLE AND HARD-WEARING**, cement is used to lay bricks, plaster walls, and construct roads.



Steel

Steel is an alloy (mixture) of iron and other substances. It has been used for **thousands of years**, but it really took off in 1858 when Englishman **Henry Bessemer** invented a process to produce large amounts of it very cheaply. In 1913, English laboratory researcher **Harry Brearley** was in pursuit of a long-lasting steel for gun barrels, when he discovered a type that **resisted damage** from acids. Today, his **STAINLESS STEEL** is everywhere, from items such as saucepans and sinks to tiny nuts and bolts holding household goods together.

Strong steel girders are used in construction to form the supporting structures of buildings.



Material world

People have been crafting objects since ancient times, but these marvelous materials have transformed the way we live.

The materials that BUILT the modern world

Did you know?
The ancient Romans used volcano ash to make their concrete. Today, about 1.3 billion tons of cement is produced every year.

